MedVL-SAM2: A unified 3D medical vision-language model for multimodal reasoning and prompt-driven segmentation

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Abstract— Medical vision–language models (VLMs) have shown strong potential on image-level, text-centric tasks such as report generation and visual question answering (VQA). However, achieving fine-grained visual localization and volumetric spatial reasoning in 3D medical VLMs remains challenging, particularly when attempting to unify these capabilities within a single framework. To address this challenge, we propose MedVL-SAM2, a unified 3D medical VLM that simultaneously supports report generation, VQA, and multi-paradigm segmentation, including semantic, referring, and interactive segmentation. MedVL-SAM2 integrates image-level reasoning and pixel-level localization through a cohesive architecture tailored for 3D medical imaging. The model is trained via a multi-stage pipeline. First, it is pre-trained on a large-scale collection of 3D CT image–text pairs to align volumetric visual representations with radiology language embeddings. It is then jointly optimized using both language understanding and segmentation objectives on a comprehensive 3D CT segmentation dataset. This joint training enables flexible interaction through language, point, or box prompts, effectively unifying high-level semantic reasoning with spatially precise localization. Experiments demonstrate that MedVL-SAM2 achieves state-of-the-art performance across report generation, VQA, and diverse 3D segmentation tasks. These results suggest that high-level semantic reasoning and precise volumetric localization can be effectively integrated within a unified 3D medical VLM framework. Code is publicly available at https://github.com/astlian9/MedVL-SAM2.

Index Terms— Vision Language Model, Image Segmentation, Report Generation, Visual Question Answering

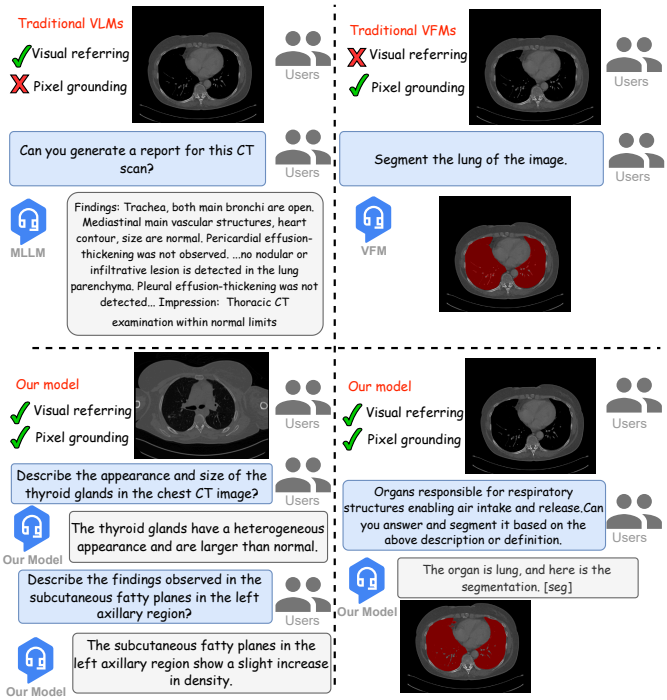

Fig. 1. Comparison of the proposed model to traditional text-response-only medical vision-language models (VLMs) and mask-response-only medical vision foundation models (VFMs).

I. Introduction

LEVERAGING large-scale radiology image-text pairs, medical vision-language models (VLMs) exhibit strong reasoning ability over both global image semantics and clinical narratives. Furthermore, several 2D VLM studies such as MIMO [1] and MedPLIB [2] have extended VLMs with spatial localization capabilities, enabling them to identify disease-related regions or anatomical structures referenced in language prompts. However, the reliance on 2D vision-language architectures inherently constrains their effectiveness in 3D medical image analysis. Specifically, 2D VLMs lack volumetric perception and spatial consistency across slices, both of which are essential for understanding complex 3D structures such as tumors, vessels, and organs. Consequently, while 2D VLMs may identify features within an individual slice, it fails to capture the spatial continuity, anatomical context, and morphological variations present across the full 3D volume.

Recent efforts have begun to explore 3D medical VLMs, such as Med3DVLM [3] and CT-CHAT [4], which process volumetric data and generate textual outputs. Nevertheless, these models remain limited to language-only output, lacking voxel-level localization or segmentation

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capabilities, which restricts their applicability in clinically meaningful interpretation and interaction. A significant gap thus remains in developing 3D medical VLMs capable of fine-grained spatial localization and aligning textual concepts with 3D spatial regions. To address this, several 3D VLMs with segmentation abilities have been proposed, including M3D [5] and VILA-M3 [6]. M3D incorporates 3D information but is constrained by the low input resolution of its vision encoder and the design of its projection layer, introducing substantial volumetric information loss. Meanwhile, VILA-M3 relies on domain-specific expert modules for downstream tasks, and its performance may become unstable when these components are not effectively activated. These limitations underscore the need for a unified 3D medical VLM capable of understanding volumetric image structures while preserving strong language-vision alignment.

To address these challenges, we propose MedVL-SAM2, a unified 3D medical VLM that extends a LLaVA-style architecture to incorporate volumetric perception and multimodal reasoning. Fig. 1 illustrates the differences between our model and traditional medical VLMs and vision foundation models (VFMs). MedVL-SAM2 incorporates image-level understanding and pixel-level segmentation within a single framework, enabling both high-level language-driven analysis and fine-grained spatial localization. Specifically, our model integrates the prompt-driven paradigm of SAM2 into a 3D vision-language backbone, supporting multimodal interaction through language, point, and bounding-box prompts. Our key contributions are summarized as follows:

- The proposed model provides a unified 3D medical VLM capable of jointly performing image-level reasoning and pixel-level perception, supporting report generation, VQA, and multiple types of 3D segmentation.
- The framework incorporates multimodal prompting through language, point, and bounding-box inputs, enabling flexible and interactive 3D image segmentation.
- The method achieves superior performance on two large-scale benchmarks, consistently surpassing current approaches in both reasoning and segmentation tasks.

II. Related Works

A. VLMs

VLMs integrate visual encoders with large language models (LLMs) for vision-language understanding. Early approaches rely on complex fusion mechanisms, such as cross-attention (Flamingo [11]) and query transformers (BLIP-2 [12]), while more recent approaches have converged toward simpler, unified designs. LLaVA [13], [14] shows that a vision encoder with a linear projection is sufficient for effective vision-language alignment, leading to numerous extensions (LLaVA-Next [15], VILA [16], InternVL [17]), and recent unified foundation models (GPT-5 [18], Gemini2.5 [19], Qwen3 [20]). While these models excel in text generation and high-level reasoning, they lack inherent spatial localization capabilities. Recent works have integrated pixel-level localization through segmentation adapters (LISA [21], VISA [22]), region encoders (GLaMM [23], Ferret [24], [25]), and multi-prompt frameworks (SA2VA [26]). However, these VLMs operate solely on 2D images or videos and cannot process 3D volumetric medical data.

TABLE I
CAPABILITY COMPARISON BETWEEN OUR MODEL AND CURRENT STATE-OF-THE-ART METHODS ACROSS REPORT GENERATION (RG), VISUAL QUESTION ANSWERING (VQA), SEMANTIC SEGMENTATION (SS), REFERRING SEGMENTATION (RS), AND INTERACTIVE SEGMENTATION (IS).

Method	RG	VQA	SS	RS	IS
BiomedParse [7]	×	×	✓	✓	×
medSAM2 [8]	×	×	✓	×	✓
CT-CHAT [4]	✓	✓	×	×	×
Med3DVLM [3]	✓	✓	×	×	×
Med-2E3 [9]	✓	✓	×	×	×
MS-VLM [10]	✓	✓	×	×	×
M3D [5]	✓	✓	✓	✓	×
Ours	✓	✓	✓	✓	✓

B. Medical VLMs

The integration of multimodal reasoning into medical imaging has led to rapid progress in medical VLMs for report generation and VQA. Early efforts such as LLaVA-Med [27] and Med-Flamingo [28] adapt general-purpose multimodal frameworks to radiology datasets, enabling report generation and VQA. Subsequently, HuatuoGPT-Vision [29], HealthGPT [30], MedTrinity [31], and RadFM [32] extend this line of work by coupling visual-language reasoning with radiology-specific representations to improve clinical understanding. More recent models, including MIMO [1] and MedPLIB [2], improve fine-grained anatomy detection through language-guided segmentation. However, all these models are constrained to 2D slice-based analysis and cannot capture volumetric spatial continuity, which is essential for accurate 3D medical reasoning. To address this limitation, several models have advanced toward volumetric understanding. CT-CHAT [4] and Med3DVLM [3] align 3D CT embeddings with radiology reports for volumetric text generation but lack segmentation capabilities. VividMed [33], MS-VLM [10] and Med-2E3 [9] extend the CT-CHAT model by integrating both 3D and 2D features, but remain restricted to text-only tasks. M3D [5] introduces segmentation heads for spatial reasoning but suffers from information loss caused by the compression of input image dimension and projection layer, while VILA-M3 [6] employs an agent-triggered segmentation mechanism that highly depends on explicit token activation.

C. Vision Foundation Models

The Segment Anything Model (SAM) [34] and SAM2 [35] establish universal segmentation capabilities

through multi-prompt interaction (points and boxes). Their strong zero-shot generalization abilities have motivated adaptations to the medical domain [36], [37]: MedSAM [38] and MedSAM2 [8] fine-tune SAM architectures on radiology datasets, while SAM-Med3D [39], SegVol [40], and parameter-efficient adapters [41] extend these models to 3D volumetric data. Additional efforts such as UniSeg [42] and BiomedParse [7] further broaden biomedical segmentation across multiple imaging modalities. Despite their strong perceptual capability, these VFMs remain segmentation-oriented and lack language-guided reasoning. Our framework addresses this gap by integrating SAM2's promptable segmentation within a LLaVA-style 3D medical VLM backbone, combining precise mask generation with semantic understanding to enable unified 3D volumetric reasoning and interactive segmentation. A comparison of task coverage across existing SOTA 3D medical VLMs, VFMs, and our model is provided in Table. I.

III. Methods

A. Problem definition

3D medical vision-language tasks cover a broad spectrum, including report generation, VQA, referring segmentation, and interactive segmentation. To support all these tasks within a single model, we introduce a unified formulation that accommodates both text-centric and mask-centric predictions. Given a 3D medical volume $\mathcal{I}_{in} \in \mathbb{R}^{X\times Y\times Z}$ with size of $X \times Y \times Z$, a sequence of text tokens $\mathcal{T}_{in} \in \mathbb{R}^{N\times D}$ with size of $N \times D$, and a set of prompt tokens $\mathcal{P}_{in}$, the unified multi-task problem can be formulated as

$$(\mathcal{T}_{out}, \mathcal{M}_{out}) = \mathcal{F}\left(\Theta; \mathcal{I}_{in}, \mathcal{T}_{in}, \mathcal{P}_{in}\right), \tag{1}$$

where $\mathcal{F}(\cdot)$ denotes the unified 3D medical VLM, $\Theta = \{\theta_{vlb}, \theta_{seg}\}$ denotes the trainable parameters including the vision-language backbone parameters θ_{vlb} and the segmentation module parameters θ_{seg}, $\mathcal{T}_{out}$ represents the output text tokens, and $\mathcal{M}_{out}$ is the output segmentation mask. The output text tokens $\mathcal{T}_{out}$ are decoded by the tokenizer to generate the text output. Different from the current unified medical VLMs, our method provides visual prompts $\mathcal{P}_{in}$ to enhance interactive segmentation. Here $\mathcal{P}_{in}$ includes the bounding box prompt $\mathcal{P}_{bbx}$ and the point prompt $\mathcal{P}_{pt}$.

B. Architecture Overview

As shown in Fig. 2, our approach consists of a LLaVA-like VLM and a promptable segmentation module. The VLM comprises the following components: (i) a volumetric vision encoder, (ii) an adaptive token projection layer, and (iii) an LLM backbone with LoRA layers. Our architecture addresses three key challenges in unified 3D medical VLM design: Challenge 1: Volumetric Understanding. Unlike 2D VLMs, 3D medical imaging requires capturing spatial relationships across slices. We address this through a 3D-aware vision encoder (M3D-CLIP) that preserves volumetric context. Challenge 2: Computational Efficiency. Direct encoding of 3D volumes produces 2048 visual tokens per image (fourfold more than 2D LLaVA), causing prohibitive LLM inference costs. We address this via a MLP-Mixer projection layer that reduces tokens to 512 while preserving spatial information. Challenge 3: Cross-Slice Consistency. Unlike 2D segmentation, where each image is independent, 3D segmentation requires anatomical continuity across slices. To alleviate this issue, the proposed model leverages the memory attention mechanism of SAM2 to maintain slice-wise consistency throughout the volume.

C. Vision-Language Processing Pipeline

Volumetric vision encoder. Following the design of LLaVA, a CLIP-based 3D encoder [43] is adopted as the vision encoder. Given an input 3D volume $\mathcal{I}_{in}$, the image embeddings v_{in} are derived via a 3D CLIP-based encoder E_{vc} resulting in $v_{in} = E_{vc}(\mathcal{I}_{in}) \in \mathbb{R}^{n\times d}$, where n is the number of visual tokens and d is the token dimension. In our framework, model weights from the pre-trained 3D vision-transformer (3D ViT) [5] are utilized.

Adaptive token projection layer. The high spatial dimensionality of 3D medical images leads to very large token sequences after ViT patch embedding. In 3D CLIP, a $32 \times 256 \times 256$ volume is partitioned into $8 \times 16 \times 16$ patches, producing 2048 visual tokens, which results in a four-fold increase in inference time compared with the 512-token LLaVA model. Furthermore, because anatomical structures occupy only a small portion of the volume, most visual tokens correspond to background regions, making direct processing of all tokens highly inefficient.

To address this issue, we introduce two MLP-Mixer layers [44] as a projector that compresses the visual tokens from 2,048 to 512 with minimal information loss. Each MLP-Mixer layer contains two fully connected sublayers with nonlinearities applied independently to each row of the input tensor: the first performs channel mixing, and the second performs token mixing. Omitting layer indices, the mixer layer is formulated as

$$\begin{aligned} U^T &= W_2\sigma(W_1\ \mathrm{LayerNorm}(v_{in}^T)), \\ T_v &= W_4\sigma(W_3\ \mathrm{LayerNorm}(U)), \end{aligned} \tag{2}$$

where σ denotes the GELU activation function, $U \in \mathbb{R}^{\hat{n}\times d}$ and $T_v \in \mathbb{R}^{\hat{n}\times \hat{d}}$ are the visual tokens aligned with the LLM's text embedding dimensions ($\hat{n} = 512$ and $\hat{d} = 2048$ in our case). The computational complexity of the MLP-Mixer layer is linear with respect to the number of input patches, which enables efficient compression of 3D volumetric embeddings while preserving spatial information.

LLM backbone with LoRA. In our model, InternVL-2.5-4B [17] is utilized as the LLM backbone for its strong multimodal understanding and text generation capabilities. LoRA [45] layers are applied to the query, key, and value projection matrices in all transformer layers to enable efficient fine-tuning. The overall process is formulated as: $T_{out} = LLM(T_{in} + T_v)$, where T_{in} denotes the tokenized

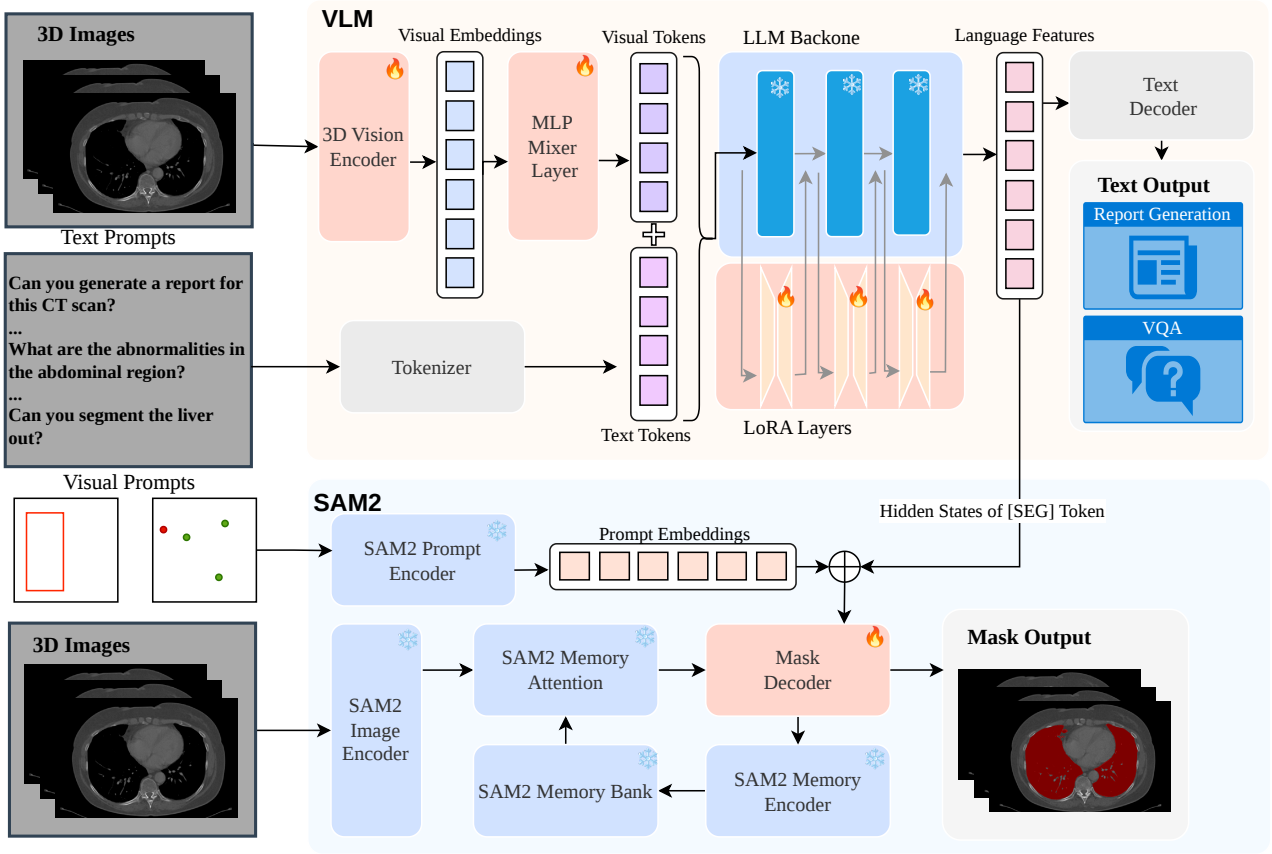

Fig. 2. Overview of the proposed architecture, which integrates a LLaVA-style VLM with a SAM2-based segmentation module. The VLM processes 3D volumes and generates text autoregressively. When a [SEG] token is produced, its hidden state is extracted and fed into SAM2's prompt encoder, where it is fused with optional visual prompts (points or boxes) to generate the final segmentation mask.

text prompt and T_v denotes the visual tokens produced by the projection layer.

D. [SEG] Token Controlled Segmentation

In our framework, SAM2 is employed as the segmentation module for its strong performance and native support for multi-prompt interaction. The 3D medical volume is processed slice-by-slice using memory attention to maintain cross-slice consistency. Voxel-level segmentation is enabled through a hidden state of [SEG] token, denoted as $[\text{SEG}]_{hs}$, which serves as a semantic prompt and complements geometric prompts such as points and bounding boxes. Specifically, $[\text{SEG}]_{hs}$ is passed to the SAM2 decoder, where it is concatenated with prompt embeddings from the SAM2 prompt encoder and decoded into segmentation masks. During training, the SAM2 decoder learns to interpret the spatial-temporal prompt, while gradients flowing through the [SEG] token enable the VLM to learn how to use it for accurate localization. The segmentation module can be denoted as $\mathcal{M}_{out} = SAM2(\mathcal{I}_{in}, [\text{SEG}]_{hs}, \mathcal{P}_{in})$. A unidirectional design is adopted in which SAM2 outputs are not fed back into the LLM, allowing the architecture to remain simple, minimizing cross-modal alignment overhead, and enabling independent module upgrades.

IV. Implementation Details

A. Datasets

CT-RATE. To evaluate report generation and VQA performance, we utilize the CT-RATE dataset [4], a large-scale, public resource comprising 21,304 3D non-contrast chest CT volumes paired with corresponding free-text radiology reports and over 2.7 million VQA pairs. The dataset is officially partitioned into a training set of 20,000 patients and a test set of 1,304 patients. For report generation, the "Findings" and "Impression" sections are extracted from the original radiology reports and used as ground truth. A comprehensive instruction-tuning dataset derived from the CT-RATE reports is employed for VQA, which is specifically structured to cover three distinct question-answer formats, including short answers, long answers, and multiple-choice questions, to evaluate the robustness of the model's interactive and diagnostic reasoning capabilities. M3D-Seg. The M3D-Seg dataset [5] is used to evaluate 3D segmentation performance. M3D-Seg is a unified benchmark that aggregates 3D CT scans from 25 public segmentation datasets, providing 5,772 labeled volumes paired with 119,267 category text descriptions and segmentation masks covering 211 anatomical structures. For interactive segmentation, we generate point and bounding-box prompts following the MedSAM2 protocol:

TABLE II
MODULE PARAMETERS AND MODEL SIZE

Module	Parameters	Size
Vision Encoder	198.05M	377.76 MB
Projection Layer	7.09M	13.52 MB
LLM Backbone	3.39B	6.33 GB
Segmentation Module	224.43M	428.07 MB
Total	3.82B	7.15 GB

TABLE III
PERFORMANCE COMPARISON ON REPORT GENERATION.

Method	BLEU-1	ROUGE	METEOR	CIDEr	BERTScore
LLaVA [13]	18.47	10.34	11.22	0.050	30.55
CXR-LLaVA [47]	9.12	5.05	2.86	0.049	39.18
LLaVA-Med [27]	9.01	5.24	2.24	0.006	40.65
M3D [5]	14.49	19.25	14.11	0.081	84.12
CT-CHAT [4]	43.64	31.57	23.06	0.221	88.12
Ours	41.85	34.59	22.15	0.237	89.33

three positive points are randomly sampled within the target region, and boxes are derived from the ground-truth masks with ±5% spatial jitter to simulate imperfect user input.

B. Multi-Stage Training

Inspired by Cambrian-1 [46] framework, we employ a three-stage progressive training strategy to stably align 3D visual features, language representations, and voxel-level localization capability. In the first stage, only the MLP-Mixer projection layer is optimized, while the vision encoder, LLM, and SAM2 modules remain frozen. This stage establishes the initial vision-language alignment and is trained for 3 epochs on the report-generation subset of CT-RATE using the autoregressive language modeling loss $\mathcal{L}_{text}$ [26]. In the second stage, the trainable set is expanded to include the vision encoder, the projection layer, and the LLM LoRA parameters, with SAM2 still frozen. This stage focuses on strengthening cross-modal representation learning and is trained for 5 epochs on CT-RATE, sampling report-generation, and VQA tasks with equal probability under the same objective $\mathcal{L}_{text}$. In the final stage, the SAM2 decoder is unfrozen to enable full multimodal spatial localization and segmentation. Joint optimization is performed for 5 epochs using both the CT-RATE and the M3D-Seg datasets using the composite objective function

$$\mathcal{L}_{joint} = \lambda_{text}\mathcal{L}_{text} + \lambda_{mask}\left(\mathcal{L}_{CE} + \lambda_{Dice}\mathcal{L}_{Dice}\right), \quad (3)$$

where $\mathcal{L}_{CE}$ is the voxel-wise binary cross-entropy loss, $\mathcal{L}_{Dice}$ denotes the Dice loss. Coefficients λ_{text}, λ_{mask}, and λ_{Dice} control the relative contributions of each term. This staged training schedule reduces catastrophic forgetting, stabilizes optimization, and results in a unified model capable of robust image-text alignment and mask prediction. Detailed module parameters of the proposed method are presented in Table II. Eight NVIDIA B200 GPUs were used for training and one B200 GPU was used for evaluation. For training Stages 1 and 2, where the trainable parameters did not include SAM2 modules, the batch size was set to 16 per GPU (effective batch size 128). Training Stage 1 for vision-language alignment required approximately 6 hours over 3 epochs, while training Stage 2, which strengthened language reasoning, required approximately 18 hours over 5 epochs. For training Stage 3 involving joint-training, the batch size was set to 1 per GPU with 8-step gradient accumulation (effective batch size 8). The entire joint-training process required approximately 110 hours over 10 epochs. Loss function hyperparameters for Stage 3 were set as follows: $\lambda_{text} = 0.5$, $\lambda_{mask} = 2.0$, and $\lambda_{Dice} = 1.0$. These values were determined through validation set tuning.

V. EXPERIMENTS

A. Evaluation on Report Generation

We evaluate report generation on the CT-RATE test set, which contains 1,304 CT-report pairs. Traditional n-gram-based NLP metrics and LLM-based metrics are used for assessment, including BLEU [48], ROUGE [49], METEOR [50], CIDEr [51], and BERTScore [52]. While n-gram metrics capture lexical overlap, they offer limited semantic insight. In contrast, LLM-based metrics assess a deeper semantic similarity between the generated reports and the ground-truth findings. We compare our model with representative multimodal baselines, including LLaVA [13], CXR-LLaVA [47], LLaVA-Med [27], M3D [5], and CT-CHAT [4]. All models are evaluated under identical inference conditions. The results are presented in Table. III and qualitative examples are shown in Fig. 3. Different aspects of the generated reports are color-coded for clarity: clinical findings in orange, COVID-19 related findings in blue, normal/healthy findings in green, and diagnostic conclusions in red. Our model achieves strong performance across both traditional and LLM-based evaluation metrics. Specifically, it outperforms 2D medical multimodal models such as LLaVA, CXR-LLaVA, and LLaVA-Med by large margins, indicating better textual fluency and clinical content accuracy. Compared with 3D baselines, our model significantly surpasses M3D and achieves comparable or higher scores than the current state-of-the-art CT-CHAT, demonstrating improved semantic fidelity and reasoning ability in the volumetric report generation task. Task-specific templates were used for report generation task. Representative templates are provided in Appendix A.

B. Evaluation on VQA

We evaluated VQA performance on the CT-RATE dataset, which includes three subtasks: long-answer, short-answer, and multiple-choice VQA. Qualitative examples are shown in Fig. 4. The long-answer and short-answer subtasks are treated as open-ended VQA, where the model generates free-form responses, while the multiple-choice subtask is treated as closed-ended VQA, where the model selects one answer from predefined options. For

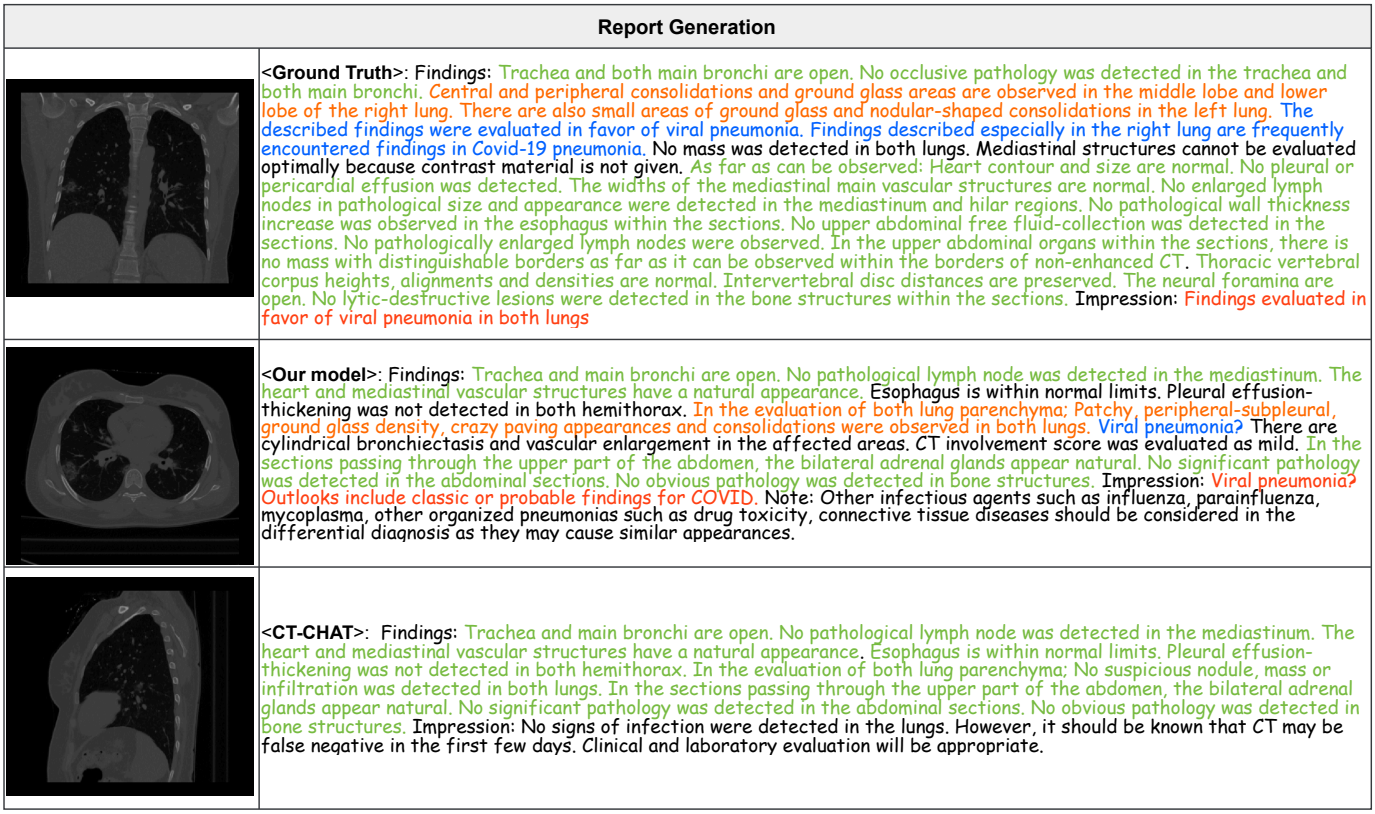

Fig. 3. Qualitative comparison of the proposed method with CT-CHAT on the report generation task. Coronal, axial, and sagittal views of the corresponding CT scans are shown. Different aspects of the reports are color-coded for improved readability.

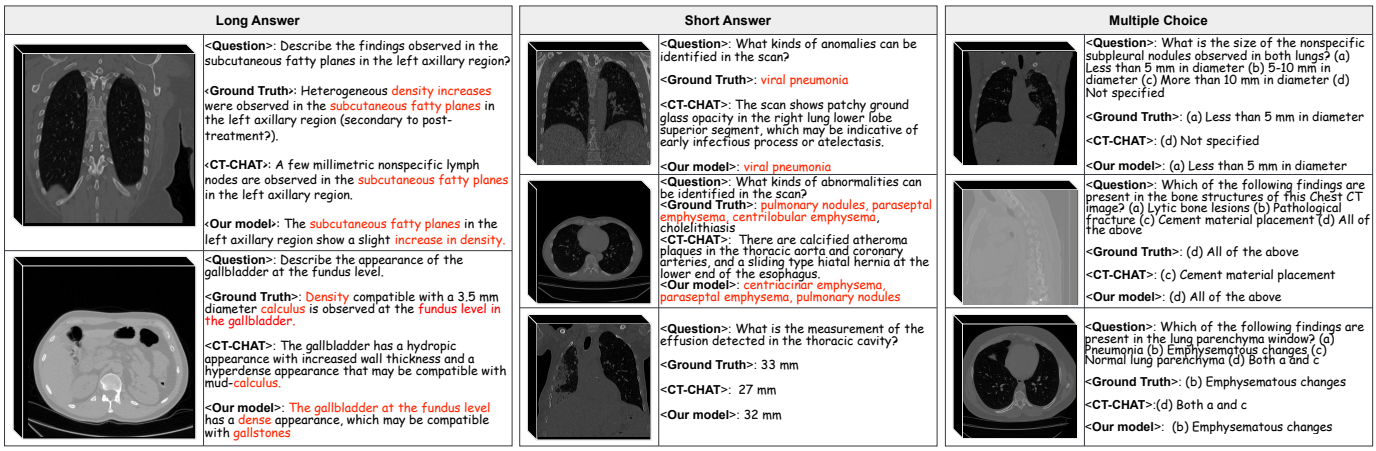

Fig. 4. Comparison between the proposed methods to CT-CHAT on VQA tasks, including Long Answer, Short Answer, and Multiple choice subsets. Critical clinical information was highlighted in red. For CT-CHAT, the input question additionally contained a special token ([short_answer], [long_answer], [multiple_choice]) to indicate the specific subset to evaluate.

open-ended tasks, we employ BLEU, ROUGE, METEOR, CIDEr, and LLM score to evaluate lexical and semantic similarity between generated and reference answers. For the closed-ended task, accuracy is used as the measure. Based on the findings from the report-generation experiments, which reveal that 2D medical multimodal models such as LLaVA, CXR-LLaVA, and LLaVA-Med, and the 3D multimodal model M3D exhibit substantial performance limitations, we focus our comparison on the current state-of-the-art CT-CHAT model. Quantitative results are presented in Table. IV. Our model achieves comparable or superior performance across all VQA subtasks. It delivers consistent improvements in ROUGE, LLM score, and accuracy, indicating stronger semantic alignment and reasoning capabilities. Notably, our model achieves 89.74% accuracy on multiple-choice VQA, surpassing CT-CHAT by a clear margin. These results confirm that incorporating prompt-driven segmentation and volumetric understanding enhances multimodal reasoning and overall comprehension in both open-ended and closed-ended 3D

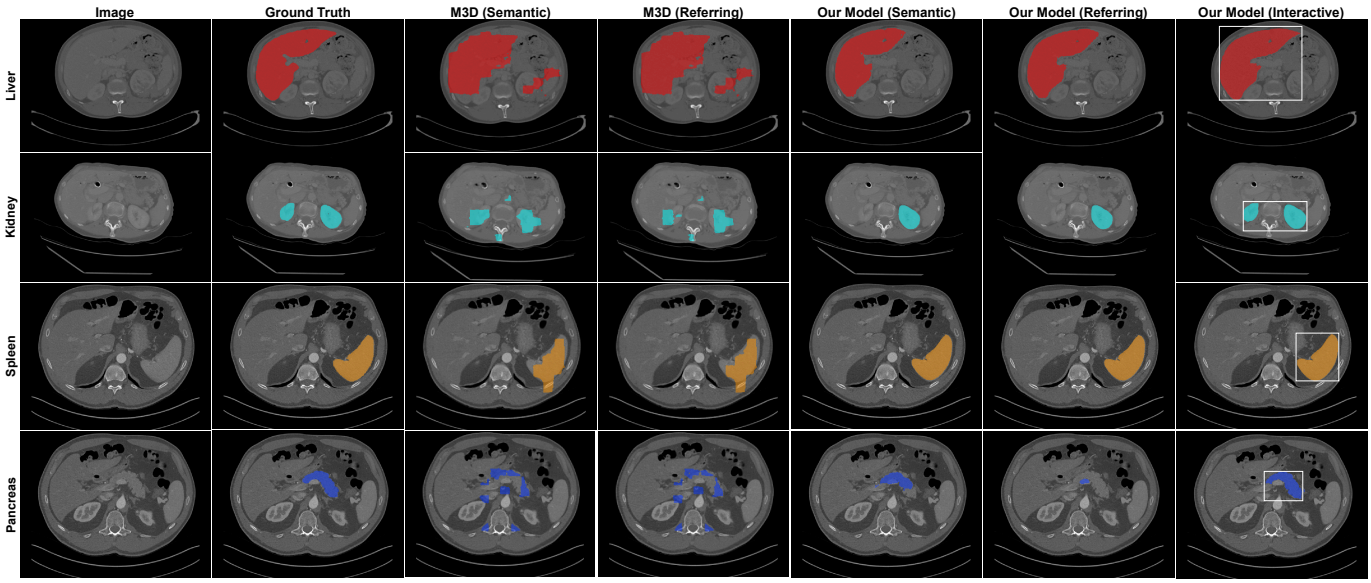

Fig. 5. Qualitative comparison of the proposed method with M3D on referring and semantic segmentation. Interactive segmentation results using bounding-box prompts are also included for comparison (ACT-1K dataset). The liver is shown in red, the kidney in cyan, the spleen in orange, and the pancreas in blue.

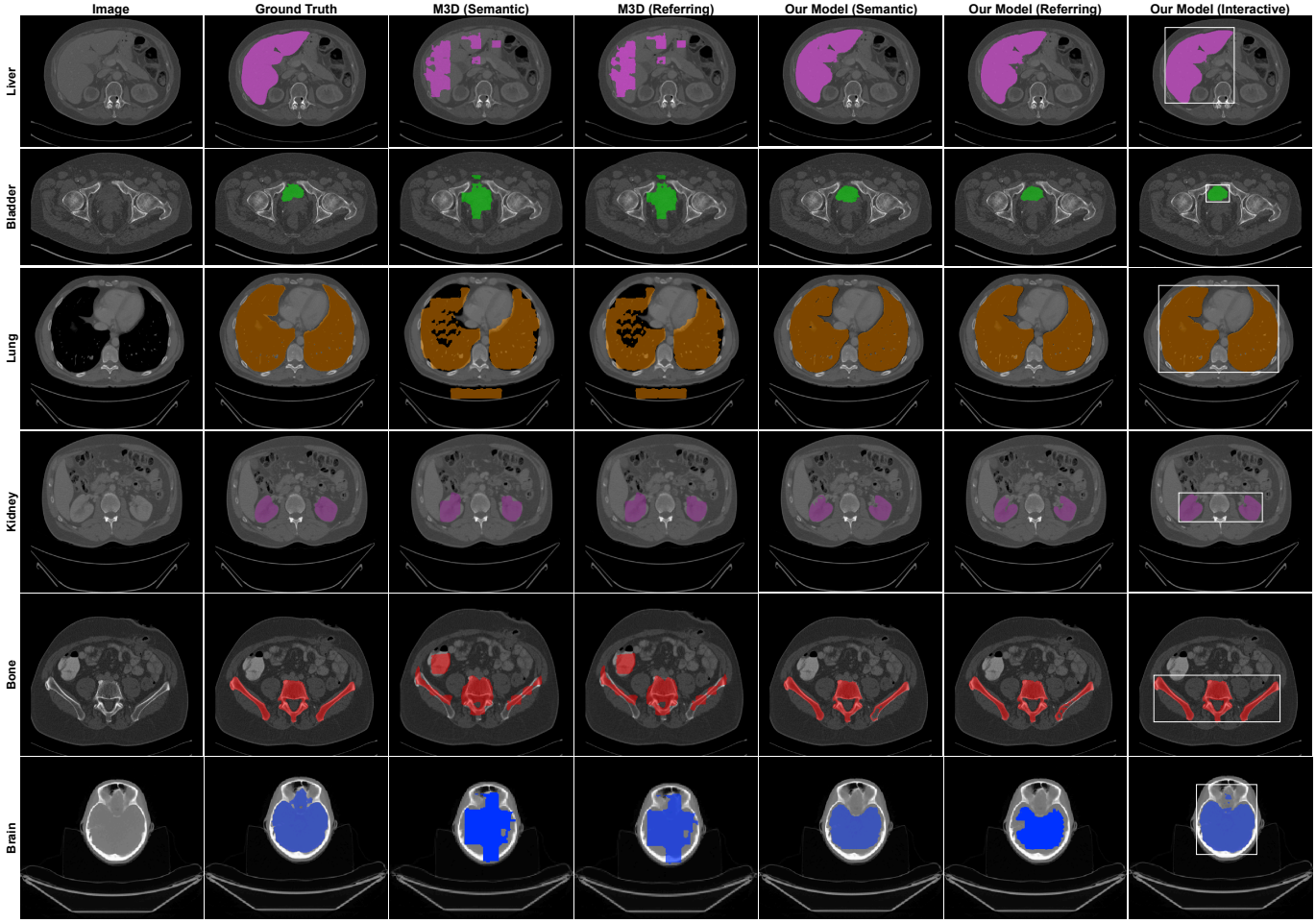

Fig. 6. Comparison of the proposed method with M3D on referring and semantic segmentation. Interactive segmentation results using bounding-box prompts are also included for comparison (CTOrg dataset shown). Liver is shown in magenta, bladder in green, lung in brown, kidney in purple, bone in red, and brain in blue.

medical VQA tasks.

C. Evaluation on Referring Segmentation

We evaluated referring segmentation on the M3D-Seg test set using the same data split as the M3D model. Since no other existing models support referring segmentation for 3D medical volumes, we use only SegVol and M3D as baseline methods. Performance is reported on three benchmark subsets of M3D-Seg dataset, including CTOrg [53], Abdomen-CT-1K (ACT-1K) [54], and TotalSegmentator (TS) [55], using the Dice coefficient as the evaluation metric. For referring segmentation, the text prompt is a randomly selected free-form description of the target region that excludes explicit class names, allowing us to evaluate the model's ability to perform language-driven spatial reasoning. To further assess the reasoning capabilities of our model, we additionally evaluate performance on semantic segmentation, where the text prompt explicitly specifies the anatomical class (e.g., "segment the liver"). Task-specific templates were used for referring segmentation and semantic segmentation tasks. Representative templates are provided in Appendix A.

Quantitative results are shown in Table. V. Across all subsets, our model achieves the highest Dice scores in both referring and semantic segmentation. Compared to SegVol, which relies solely on volumetric features, our model improves Dice by at least 10 percent for each dataset. When compared to M3D, our approach achieves consistent gains of around 5 percent, demonstrating stronger spatial reasoning and text-conditioned localization abilities. Notably, our referring segmentation performance is comparable to semantic segmentation, indicating that the model can accurately localize anatomical structures from implicit textual descriptions without explicit class labels. Overall, these results validate the effectiveness of our unified 3D multimodal framework for language-guided volumetric segmentation. The integration of prompt-driven SAM2 segmentation with multimodal reasoning enables precise and interpretable 3D spatial segmentation capability for clinical workflows involving complex 3D anatomical structures.

D. Evaluation on Interactive Segmentation

We further evaluate interactive segmentation on the same dataset and testing protocol as in the referring segmentation experiments. This setting simulates a human-in-the-loop workflow, in which users iteratively refine segmentation masks through additional prompts. Two interaction modes are examined: bounding-box prompting and point prompting. For bounding-box prompting, a random slice is selected from each 3D volume, and a bounding box with slight spatial jitter from the ground truth is provided as input. For point prompting, three randomly sampled positive points within the target region are selected on a random slice. In both settings, the model receives the same image and text inputs as in referring segmentation, augmented with the corresponding visual prompts. Dice coefficient is used as the evaluation metric. Since no existing 3D medical VLMs support interactive segmentation with text-based prompts, our semantic and referring segmentation results serve as baselines.

Quantitative results are summarized in Table. VI, and qualitative examples for both referring and interactive segmentation are shown in Fig. 5 and Fig. 6. Across all datasets, both interactive variants consistently outperform their non-interactive baselines. Bounding-box prompts yield the highest Dice scores, followed by point prompts. These findings demonstrate that incorporating user feedback substantially improves segmentation quality. We also observe that interactive refinement yielded modest improvements on CTOrg and ACT-1K. Both datasets consist primarily of large, well-defined organs (e.g., liver, lungs, kidneys), for which the language-driven [SEG] token already encodes rich spatial and semantic cues from the volumetric context. As a result, adding point or bounding-box prompts provides only incremental gains, since the model can infer coarse boundaries directly from language descriptions. In contrast, results on the TotalSegmentator dataset exhibit a larger improvement. This dataset is considerably more challenging, as it contains numerous small and complex anatomical structures, such as vertebrae, ribs, and vessels. For these fine-grained regions, the global language representation may lack sufficient spatial specificity, leading to under- or over-segmentation when relying solely on text. In these cases, explicit interactive prompts supply critical spatial priors, enabling the model to localize small anatomical regions more precisely. Overall, these findings highlight the complementary role of interactive segmentation in clinical workflows: while large structures can often be segmented reliably from language alone, small or complex anatomical targets benefit significantly from user-guided refinement, making the model more practical and adaptable for real-world clinical workflows.

VI. Ablation Studies

A. Impact of Vision Encoder and Projection Layer

To assess the contributions of the vision encoder and projection layer, we conduct ablation experiments on the CTOrg and ACT-1K subsets using the interactive segmentation task. As shown in Table. VII, two vision encoders are compared: vanilla CLIP and M3D-CLIP. For the projection layer, we evaluate three configurations: Linear, MLP, and MLP-Mixer. Replacing the vanilla CLIP encoder with the M3D CLIP backbone results in a substantial improvement in Dice scores, highlighting the importance of volumetric-aware pretraining for 3D medical data. The M3D CLIP encoder effectively captures cross-slice spatial context and anatomical continuity, while the vanilla CLIP model trained on natural images lacks such volumetric priors. Projection-layer design also has a notable impact. Linear and MLP projectors offer limited cross-modal alignment, resulting in weaker performance. In contrast, the MLP-Mixer achieves the highest Dice scores, indicating that its ability to jointly model spatial

TABLE IV
COMPARISON BETWEEN CT-CHAT AND OUR MODEL ON LONG-ANSWER, SHORT-ANSWER, AND MULTIPLE-CHOICE VQA TASKS.

Task	Long Answer					Short Answer					Multiple Choice
	BLEU	ROUGE	METEOR	CIDEr	BERTScore	BLEU	ROUGE	METEOR	CIDEr	BERTScore	Accuracy
CT-CHAT [4]	47.020	48.450	28.200	2.910	90.140	27.460	45.590	15.470	1.652	90.120	83.730
Ours	45.941	48.983	28.218	2.861	92.760	24.076	57.428	14.026	1.771	92.503	89.741

TABLE V
RESULTS ON THE REFERRING SEGMENTATION TASK.

Method	CTOrg	ACT-1K	TotalSegmentator
SegVol [40]	77.78	79.06	44.28
M3D [5] (semantic)	81.27	73.64	58.30
M3D [5] (referring)	83.49	73.63	58.50
Ours (semantic)	88.04	88.27	66.23
Ours (referring)	87.93	88.34	66.50

TABLE VI
RESULTS ON THE INTERACTIVE SEGMENTATION TASK.

Method	CTOrg	ACT-1K	TotalSegmentator
Ours (semantic)	88.04	88.27	66.23
Ours (referring)	87.93	88.34	66.50
Ours (bbox prompt)	88.76	88.71	70.51
Ours (points prompt)	88.52	88.49	69.88

and channel-wise interactions enables richer multimodal fusion and more effective volumetric reasoning. Overall, these findings validate our architectural choices: M3D-CLIP provides strong 3D semantic representations, and the MLP-Mixer projection layer facilitates robust vision-language alignment, jointly leading to significant gains in both referring and interactive segmentation tasks.

B. Impact of Multi-Stage Training Strategy

To evaluate the effectiveness of our multi-stage training approach, we compare it with an alternative one-stage end-to-end training strategy, where all components are trained simultaneously from scratch. Both strategies are evaluated on the CT-RATE dataset for the report generation task, using BLEU-1, ROUGE, METEOR, and CIDEr as evaluation metrics. Quantitative results are shown in Table. VIII. The multi-stage training strategy consistently outperforms one-stage end-to-end training across all metrics. The one-stage approach exhibits suboptimal convergence, as the simultaneous optimization of multiple large modules under competing cross-modal objectives hinders stable alignment between visual and language representations. In contrast, the progressive training schedule provides clear optimization phases, facilitating more reliable vision-language alignment. These results validate the use of a progressive multi-stage training pipeline.

TABLE VII
ABLATION STUDIES ON DIFFERENT CHOICES OF VISION ENCODERS AND PROJECTION LAYERS.

Vision Encoder	Projection Layer	CTOrg	ACT-1K
Vanilla CLIP [43]	Linear	30.18	44.65
M3D-CLIP [5]	Linear	61.99	74.58
M3D-CLIP [5]	MLP	77.78	79.83
M3D-CLIP [5]	MLP-Mixer	88.04	88.27

TABLE VIII
ABLATION STUDIES ON DIFFERENT CHOICES OF TRAINING STRATEGIES.

Training strategy	BLEU-1	ROUGE	METEOR	CIDEr
One-stage	38.58	32.53	20.89	0.2095
Multi-stage	41.85	34.59	22.15	0.2371

VII. DISCUSSION

The primary contribution of this work is the integration of image-level semantic reasoning with fine-grained volumetric perception within a unified 3D VLM. The proposed framework is capable of processing 3D medical images and supporting multiple tasks, including report generation, VQA, and various forms of 3D image segmentation. This is achieved through the integration of a prompt-driven segmentation module, a 3D vision encoder, a multimodal projector, and an LLM backbone. The unified design enables joint reasoning over volumetric image features and language, while providing spatially grounded outputs. Experimental results on the CT-RATE and M3D-Seg benchmarks indicate that the proposed framework achieves competitive performance in report generation and VQA tasks. In addition, it supports referring and interactive segmentation capabilities that are not available in existing text-centric 3D VLMs. These results suggest that incorporating segmentation-aware components into 3D VLMs may offer a practical pathway toward more interpretable and clinically useful multimodal systems.

Despite these advancements, the proposed framework has several limitations. First, the current architecture adopts a unidirectional design in which the outputs of the segmentation module (SAM2) are not fed back into the LLM. While this decoupled design simplifies implementation and enables independent optimization of each component, it limits the LLM's ability to leverage segmentation outputs for iterative reasoning or self-correction. Second, the evaluation is currently restricted to 3D CT data with a distributional discrepancy across tasks. Specifically, text-driven tasks are primarily trained and evaluated on chest CT datasets [4], whereas segmentation tasks rely on whole-body CT datasets [5]. This anatomical mismatch may introduce domain shifts that affect model stability and generalization across different body regions. Future work

will focus on incorporating bidirectional interaction mechanisms between segmentation and language components, as well as further evaluating and improving the robustness and generalizability of the proposed framework.

VIII. Conclusion

In this work, we introduce MedVL-SAM2, a unified 3D VLM that integrates SAM2's promptable segmentation with a LLaVA-style multimodal framework. The model jointly supports report generation, VQA, and semantic, referring, and interactive segmentation, bridging image-level understanding with voxel level localization and segmentation. Experiments demonstrate promising performance across multiple 3D benchmarks, confirming MedVL-SAM2's strengths in volumetric reasoning, fine-grained spatial localization, and interactive segmentation.

Appendix A. Instruction Templates Used for LLM Training

This section demonstrates task-specific prompts and templates to guide the LLM across different tasks, including report generation, semantic segmentation, and referring segmentation. Figure 7 shows prompts for report generation, while Figures 8, 9, and 10 present instruction templates for segmentation tasks.

Report Generation:

- "Can you generate the report for the following chest CT image?",
- "Please provide the radiology report for the chest CT image mentioned.",
- "I need the radiology report for the given chest CT image.",
- "Could you create a report for this chest CT scan?",
- "Would you mind generating the radiology report for the specified chest CT image?",
- "Please generate the report for the chest CT image provided.",
- "Can you produce the radiology report for the attached chest CT image?",
- "I need a detailed report for the given chest CT image.",
- "Could you write the radiology report for this chest CT scan?",
- "Please give the radiology report for the specified chest CT image.",
- "Generate radiology report for the CT.",
- "Produce the report for this CT image.",
- "Write a radiology report for the following CT scan.",
- "Create a report for this chest CT.",
- "Provide the radiology report for this CT image.",
- "Can you generate the report for the following chest CT volume?",
- "Please provide the radiology report for the chest CT volume mentioned.",
- "I need the radiology report for the given chest CT volume.",
- "Could you create a report for this chest CT volume?",
- "Would you mind generating the radiology report for the specified chest CT volume?",
- "Please generate the report for the chest CT volume provided.",
- "Can you produce the radiology report for the attached chest CT volume?",
- "I need a detailed report for the given chest CT volume.",
- "Could you write the radiology report for this chest CT volume?",
- "Please give the radiology report for the specified chest CT volume.",
- "Generate radiology report for the CT volume.",
- "Produce the report for this CT volume.",
- "Write a radiology report for the following CT volume.",
- "Create a report for this chest CT volume.",
- "Provide the radiology report for this CT volume.",
- "Can you generate the report for the following chest CT scan?",
- "Please provide the radiology report for the chest CT scan mentioned.",
- "I need the radiology report for the given chest CT scan.",
- "Could you create a report for this chest CT scan?",
- "Would you mind generating the radiology report for the specified chest CT scan?",
- "Please generate the report for the chest CT scan provided.",
- "Can you produce the radiology report for the attached chest CT scan?",
- "I need a detailed report for the given chest CT scan.",
- "Could you write the radiology report for this chest CT scan?",
- "Please give the radiology report for the specified chest CT scan.",
- "Generate radiology report for the CT scan.",
- "Produce the report for this CT scan.",
- "Write a radiology report for the following CT scan.",
- "Provide the radiology report for this CT scan."

Fig. 7. Prompt templates used for report generation tasks.

Semantic Segmentation:
Questions:

- "{} Please answer and segment.",
- "{} Please output segmentation mask and answer.",
- "{} Please answer and segment based on the above description.",
- "{} Please answer and segment based on the above definition.",
- "{} Can you answer and segment it based on the above description or definition.",
- "{} Please output segmentation mask and answer based on the above description or definition.",
- "{} Please segment accordingly.",
- "{} Please provide segmentation and answer according to it.",
- "{} Now, segment it and provide your answer.",
- "{} Please segment and provide your response.",
- "{} Can you segment it accordingly?",
- "Description: {} Please answer and segment based on the above description.",
- "Definition: {} Please answer and segment based on the above definition.",
- "Description: {} Can you answer and segment it based on the above description or definition.",
- "Definition: {} Please output segmentation mask and answer based on the above description or definition.",
- "Provided description: {} Please segment accordingly.",
- "Given definition: {} Please provide segmentation and answer according to it.",
- "The description provided is: {} Now, segment it and provide your answer.",
- "Based on the provided definition: {} Please segment and provide your response.",
- "Describing the object as: {} Can you segment it accordingly?",
- "Defining it as: {} Now, segment and provide your answer.",

Answers:

- "The target is {} and the segmentation mask is [SEG].",
- "The category is {} and the mask is [SEG].",
- "It is {}, [SEG].",
- "{}, [SEG]",
- "Identified as {}, here is the segmentation: [SEG].",
- "Categorized as {}, the segmentation is: [SEG].",
- "The class is {}, and the corresponding segmentation is: [SEG].",
- "Regarding the classification, it is {}, and the segmentation is: [SEG].",
- "Classified as {}, here's the segmentation: [SEG].",
- "The label assigned is {}, and the associated segmentation is: [SEG].",
- "Category: {}, segmentation: [SEG].",
- "It's classified as {}, with the segmentation: [SEG]."

Fig. 8. Prompt templates used for referring segmentation tasks.

Semantic Segmentation:
Questions:

- "Can you segment the {class_name} in this image?",
- "Please segment {class_name} in this image.",
- "What is {class_name} in this image? Please respond with segmentation mask.",
- "What is {class_name} in this image? Please output segmentation mask.",
- "Can you segment the {class_name} in this image",
- "Please segment {class_name} in this image",
- "What is {class_name} in this image? Please respond with segmentation mask",
- "What is {class_name} in this image? Please output segmentation mask",
- "Could you provide a segmentation mask for the {class_name} in this image?",
- "Please identify and segment the {class_name} in this image.",
- "Where is the {class_name} in this picture? Please respond with a segmentation mask.",
- "Can you highlight the {class_name} in this image with a segmentation mask?",
- "Could you provide a segmentation mask for the {class_name} in this image",
- "Please identify and segment the {class_name} in this image",
- "Where is the {class_name} in this picture? Please respond with a segmentation mask",
- "Can you highlight the {class_name} in this image with a segmentation mask"

Answers:

- "It is [SEG].",
- "Sure, [SEG].",
- "Sure, it is [SEG].",
- "Sure, the segmentation result is [SEG].",
- "[SEG]."

Fig. 9. Prompt templates used for semantic segmentation tasks.

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Term Dictionary:

Kidneys:

- "Primary organ responsible for detoxifying the blood by removing harmful substances.",
- "Produces bile, a fluid that aids in the digestion and absorption of fats.",
- "Stores and regulates glycogen, a crucial energy reserve for the body.",
- "Synthesizes proteins necessary for blood clotting and immune system function.",
- "Plays a central role in metabolism, including the breakdown of carbohydrates and fats.",
- "Large organ in the upper right abdomen with various metabolic functions.",
- "Produces bile for digestion, detoxifies blood, and stores glycogen.",
- "Essential for metabolic processes and nutrient storage.",
- "Regenerates and repairs damaged tissue.",
- "Common site for conditions such as hepatitis and cirrhosis."

Liver:

- "Primary organ responsible for detoxifying the blood by removing harmful substances.",
- "Produces bile, a fluid that aids in the digestion and absorption of fats.",
- "Stores and regulates glycogen, a crucial energy reserve for the body.",
- "Synthesizes proteins necessary for blood clotting and immune system function.",
- "Plays a central role in metabolism, including the breakdown of carbohydrates and fats.",
- "Large organ in the upper right abdomen with various metabolic functions.",
- "Produces bile for digestion, detoxifies blood, and stores glycogen.",
- "Essential for metabolic processes and nutrient storage.",
- "Regenerates and repairs damaged tissue.",
- "Common site for conditions such as hepatitis and cirrhosis.",
- "Large organ located in the upper right abdomen with various metabolic functions.",

Brain:

- "Central organ of the nervous system responsible for cognition and control of bodily functions.",
- "Complex organ housed within the skull responsible for processing sensory information and controlling body functions.",
- "Command center of the nervous system responsible for processing information and coordinating bodily activities.",
- "Highly organized mass of nerve tissue enclosed within the skull responsible for cognitive functions and regulating body processes.",

Lungs:

- "Organs responsible for respiration.",
- "Respiratory organs facilitating gas exchange.",
- "Pair of organs involved in breathing.",
- "Primary respiratory structures in the body.",
- "Organs essential for oxygen exchange.",
- "Respiratory structures enabling air intake and release.",
- "Pair of vital organs for oxygenation.",
- "Main organs for inhaling and exhaling air.",

Spleen:

- "Organ located in the upper left abdomen, part of the lymphatic system.",
- "Filters blood, removing damaged blood cells and pathogens.",
- "Stores platelets and white blood cells.",
- "Involved in immune response and blood filtration.",
- "Can be affected by conditions like splenomegaly or ruptured spleen.",
- "Organ located in the upper left abdomen, part of the lymphatic system.",
- "Filters blood, removing damaged blood cells and pathogens.",
- "Organ located in the upper left abdomen, part of the lymphatic system.",
- "Organ in the lymphatic system located in the upper left abdomen.",

Fig. 10. Anatomical descriptions used as free-form text prompts for referring segmentation tasks. These descriptions replace the {} placeholder in the templates shown in Figure 8.

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